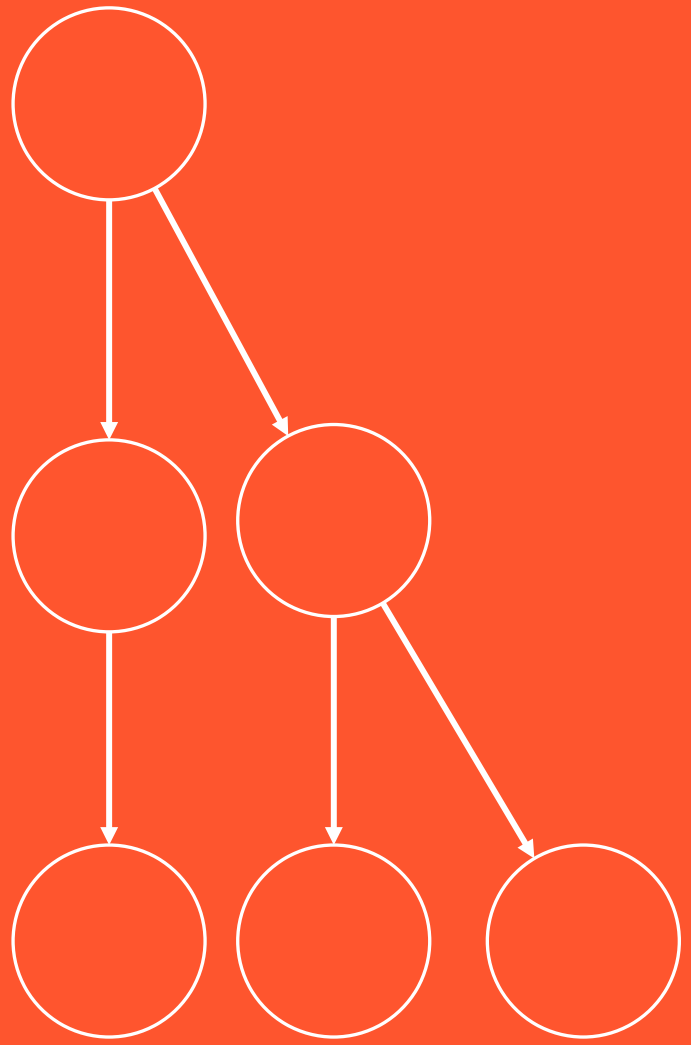


BCOG 100

Planning and Decision-making
Lecture 12.1 — Choosing Among Futures

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

What is Intelligence?

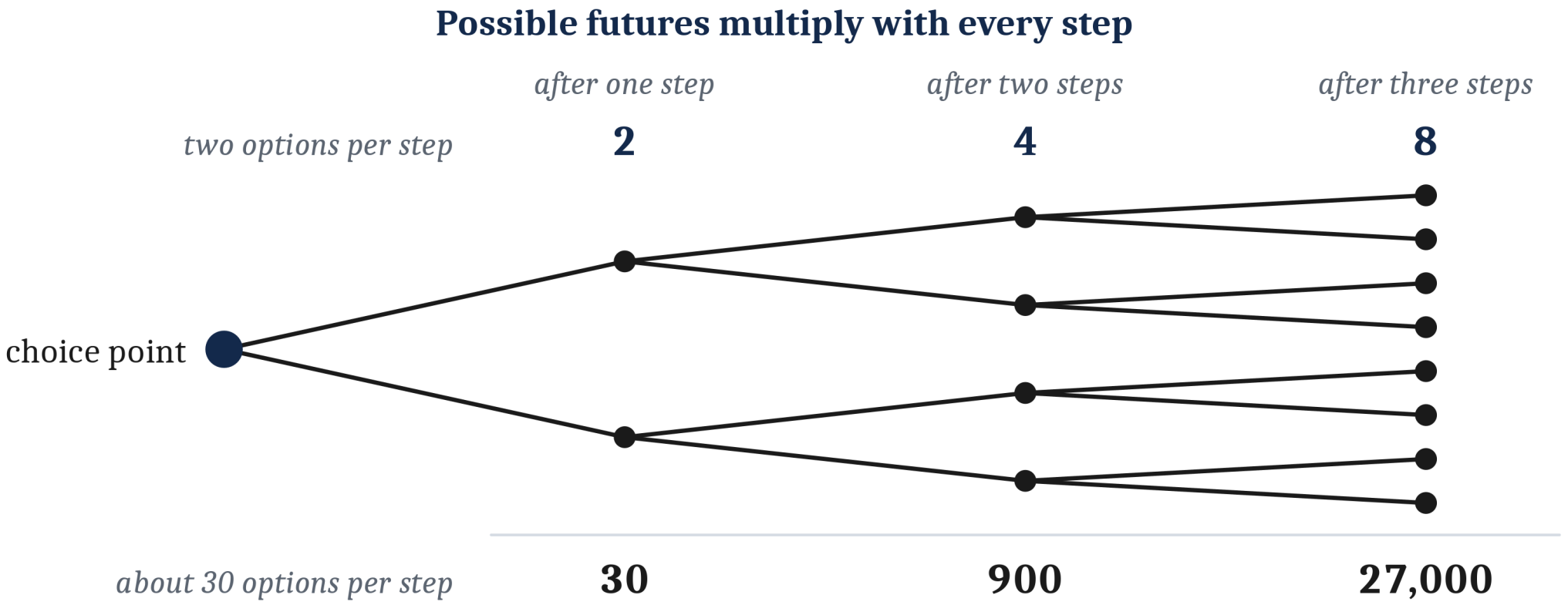
In your group, four minutes.

What is intelligence? Is answering that question an empirical question of a theoretical question?

Does the definition of intelligence matter?

Futures Multiply

Choosing among futures is hard in three ways, whoever or whatever does the choosing: the futures multiply with each step, their outcomes are uncertain, and the outcomes arrive after different delays.



counts after Huys et al. (2012)

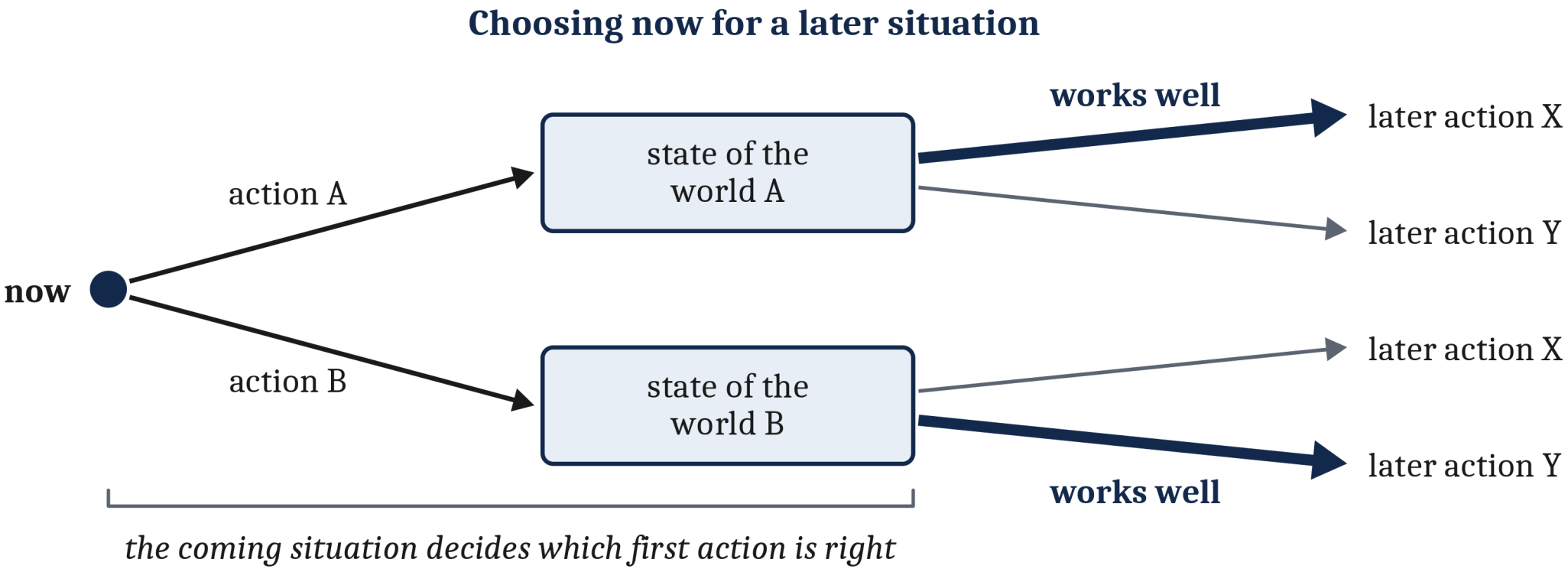
A Standard for Judging Choices

Expected value: the average of an option's outcomes, each weighted by its probability. It is a standard for judging choices, not a description of how choices are made. **Risk**: a property of a choice whose outcome is uncertain but whose probabilities are known.

Option	Possible outcomes	Expected value
Sure gain	$240, with probability 1.00	1.00 × $240 = $240
Gamble	$1,000, with probability 0.25; nothing, with probability 0.75	0.25 × $1,000 + 0.75 × $0 = $250

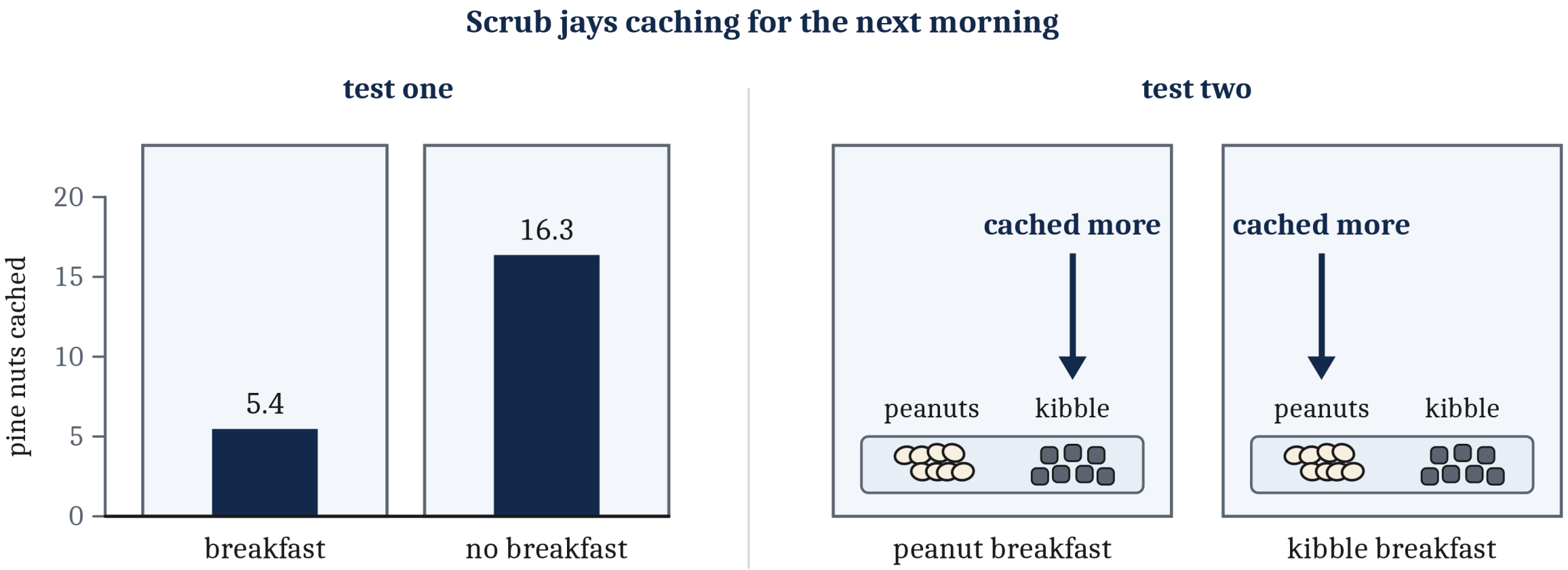
Acting Now for Later

An **affordance** is an opportunity for action that a place or object offers to a particular body. When there is more than one first action to choose between, which one is right depends on what the coming situation will call for.



Jays, Apes, and Ravens

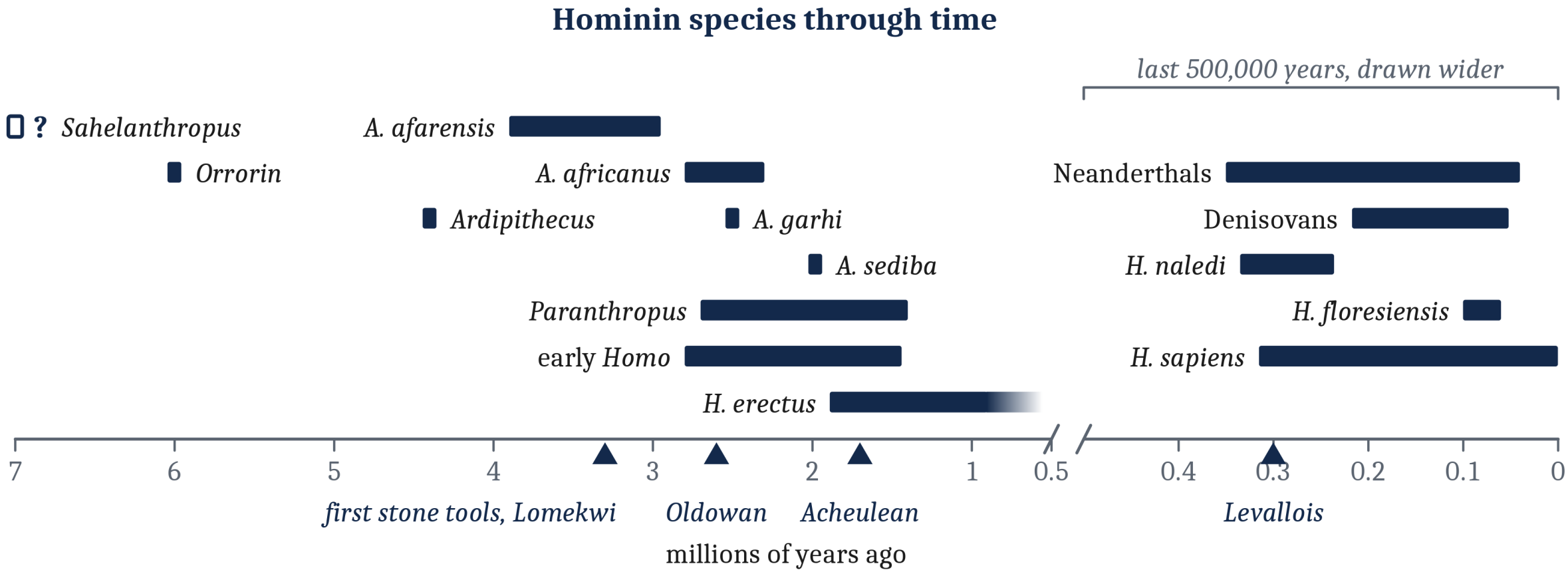
Scrub jays cached about three times as many nuts where they had gone without breakfast. These results are descriptions of behavior. They do not show whether the animal represents the later situation.



test one: averages from Raby et al. (2007), p. 919; test two: the direction of the result, p. 920

The Hominins: A Branching Tree

A **hominin** is a member of the human line after it split from the line leading to chimpanzees and bonobos. **Bipedalism**, walking habitually on two legs, came long before large brains or the first known stone tools. Later tools took longer to learn, but they are not a direct record of planning.



drawn from the sources named in the caption; a short bar marks a species known from one dated find or site

Slow Lives and Grandmothers

Life history: the timing of growth, maturity, reproduction, and death across a lifetime, in a species. **Grandmother hypothesis**: older women who gathered food for their daughters' young allowed their daughters to wean earlier and give birth again sooner, so that women who lived long after menopause left more descendants.

	Grandmother hypothesis	Hunting account
What is unusual	Long life after childbearing ends	Long life, long childhood, and a large brain together
Who supplies the extra food	Older women, gathering for their daughters' young	Mainly men, through hunting, with others contributing
Why childhood is long	A side effect of a long life	Time to learn hard foraging skills
Main evidence	Hadza grandmothers' work and grandchildren's weight	Food production by age and sex across foraging societies

What Is Intelligence?

Intelligence: how well an agent achieves its goals across a wide range of problems and environments, or the capacities and procedures that produce that achievement. **Positive manifold**: people who score well on one kind of mental test tend to score well on the others; a pattern of correlations, not a cause.

Spearman's correlations among six measures, 1904

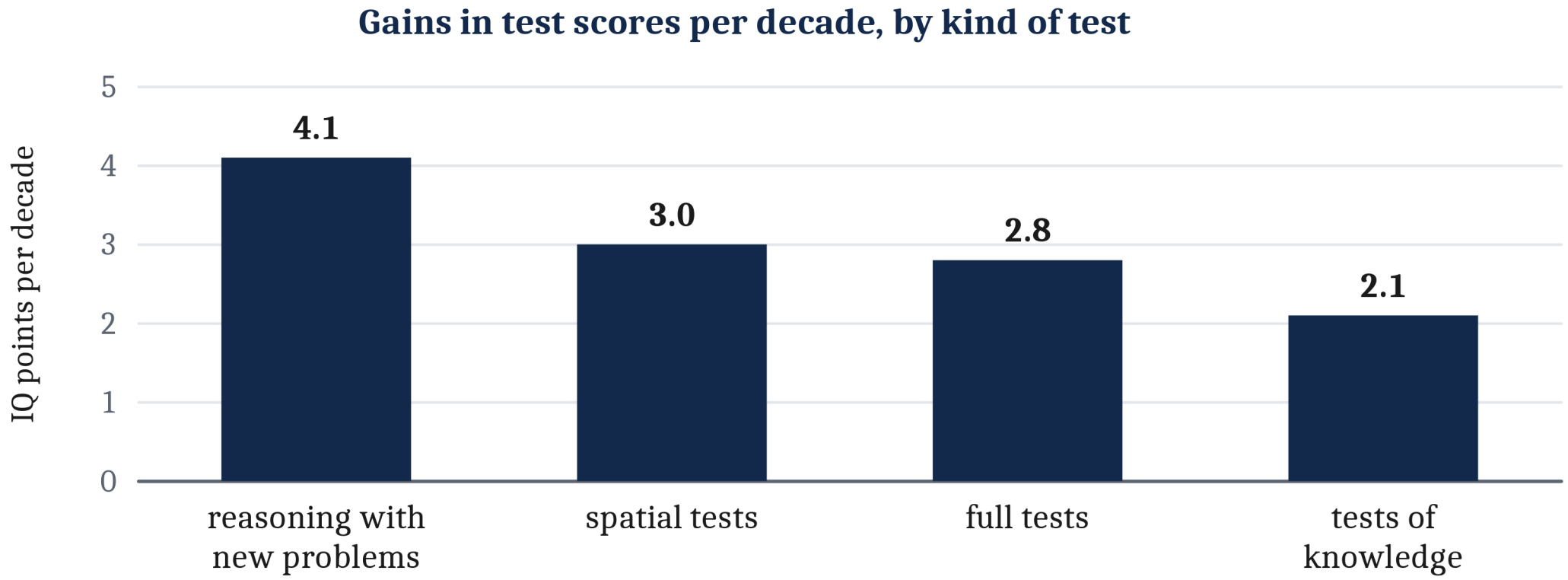
	classics	French	English	mathematics	pitch discrimination	musical talent
classics	*0.87*	0.83	0.78	0.70	0.66	0.63
French	0.83	*0.84*	0.67	0.67	0.65	0.57
English	0.78	0.67	*0.89*	0.64	0.54	0.51
mathematics	0.70	0.67	0.64	*0.88*	0.45	0.51
pitch discrimination	0.66	0.65	0.54	0.45		0.40
musical talent	0.63	0.57	0.51	0.51	0.40	

two examinations in the same subject 0.40 — 0.83

data from Spearman (1904), p. 275; about 33 boys at one school

IQ Scores and What They Predict

IQ: a score scaled so that the average in a reference sample is 100; a rank within that sample, not an amount. **Predictive validity**: how well a score predicts an outcome measured later. **Flynn effect**: the rise in average scores across the twentieth century, about 3 points per decade in the United States.

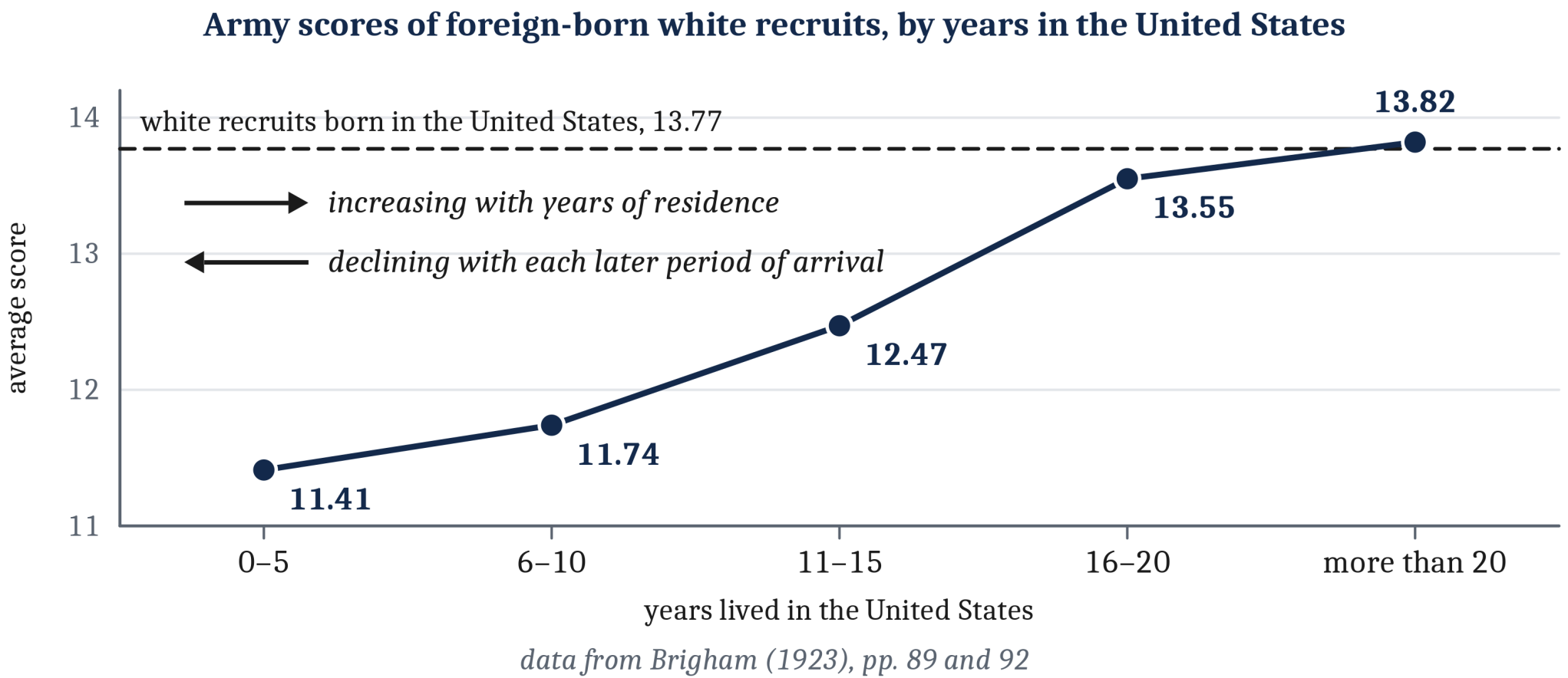


data from Pietschnig and Voracek (2015), p. 282: 271 samples, 31 countries, 1909–2013

Testing and Its Uses

Content note: eugenics, forced sterilization, and racism, with offensive language quoted from the period. **Eugenics**: the program of improving a population's quality by controlling who has children.

The definition of intelligence mattered: scores were read as if they showed inherited capacities.



Scientific Racism, and Two Claims

Scientific racism: work that presents claims of racial superiority as findings of science. The uses of the tests would have been wrong even if every claim the testers made had been true.

Claim	Kind of claim	Current evidence
Groups differ in their average scores	An observation about scores	U.S. averages have differed between groups defined by race; the difference is not explained by bias in prediction
Genes cause those differences	A claim about causes	Major reviews in 1996 and 2012 found that the direct evidence did not support a genetic explanation

To Discuss: A Hiring Test

In your group, five minutes.

A company screens job applicants with a test it says measures intelligence, because IQ scores predict performance at work.

What does an applicant's score tell the company, and what does it leave out? What else could explain why scores and job performance go together?

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